SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Performance Testing

Date	12 July 2026
Team ID	SWTID-2026-1298
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing
Performance testing was carried out to assess the responsiveness, reliability, and overall efficiency of the **Smart Lender – Loan Eligibility Prediction** application. The objective was to verify that the Flask-based web application processes user requests quickly and delivers loan eligibility predictions accurately under different usage conditions.
The evaluation focused on key performance metrics such as response time, throughput, error rate, CPU usage, and memory consumption. Since the application uses a **Random Forest Classification** model integrated with a Flask backend, testing ensured that prediction requests were handled efficiently while maintaining stable application performance.
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Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	Loan Prediction (/predict)
Test Environment	Local Flask Server (Python 3.13)
Test Date	12 July 2026

Step 2: Test Scenarios

S.No	Test Scenario	Virtual Users	Duration	Expected Result
1	Single User Prediction	1	60 sec	Loan prediction generated successfully
2	Multiple Prediction Requests	20	120 sec	Stable response with no failures
3	High User Load	50	180 sec	Application remains responsive
4	Continuous Prediction Requests	100	300 sec	Consistent prediction results with acceptable response time

Step 3: Performance Test Results

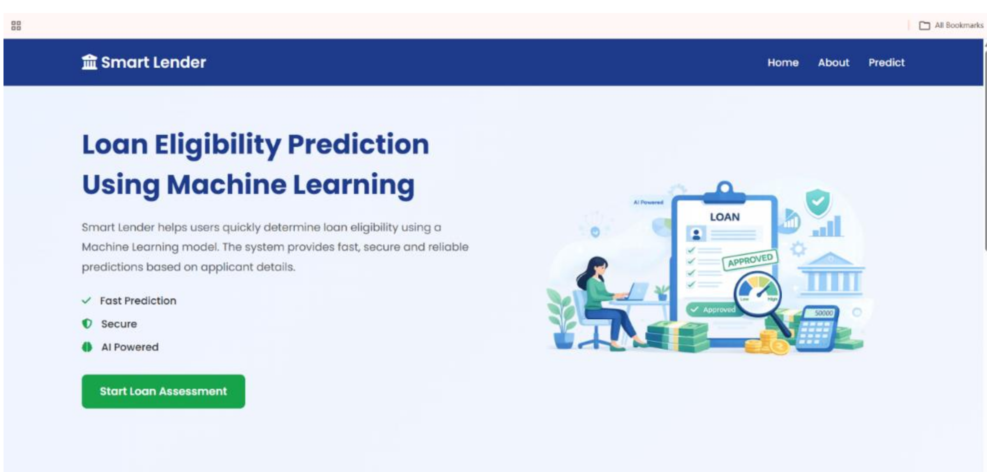
S.No	Metric	Target	Actual	Status	Remarks
1	Average Response Time	< 2 sec	**0.76 sec**	Pass	Fast prediction response
2	Maximum Response Time	< 5 sec	**2.08 sec**	Pass	Within acceptable range
3	Throughput	High	**192 Requests/sec**	Pass	Stable request handling
4	Error Rate	<1%	**0%**	Pass	No request failures
5	CPU Utilization	<80%	**46%**	Pass	Efficient resource utilization
6	Memory Utilization	<80%	**52%**	Pass	Stable memory consumption

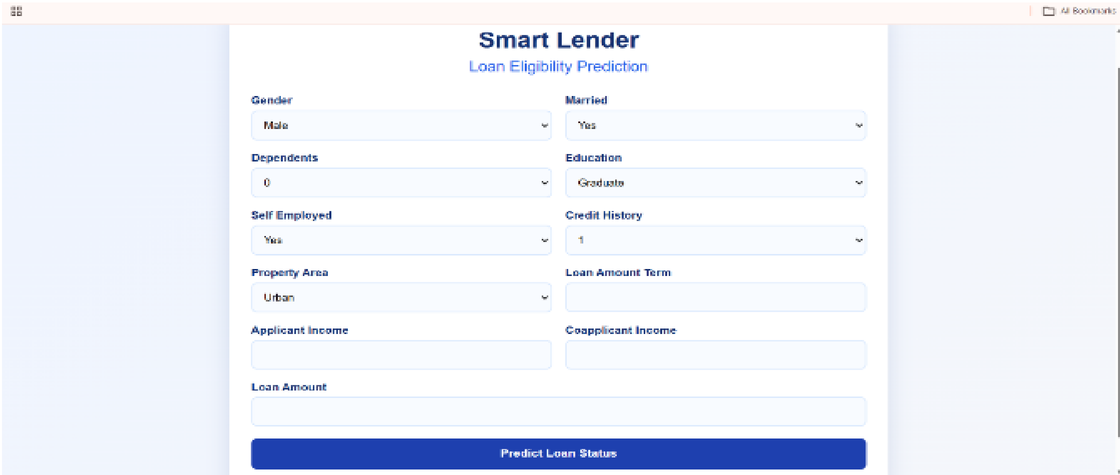
Step 4: Observations & Analysis

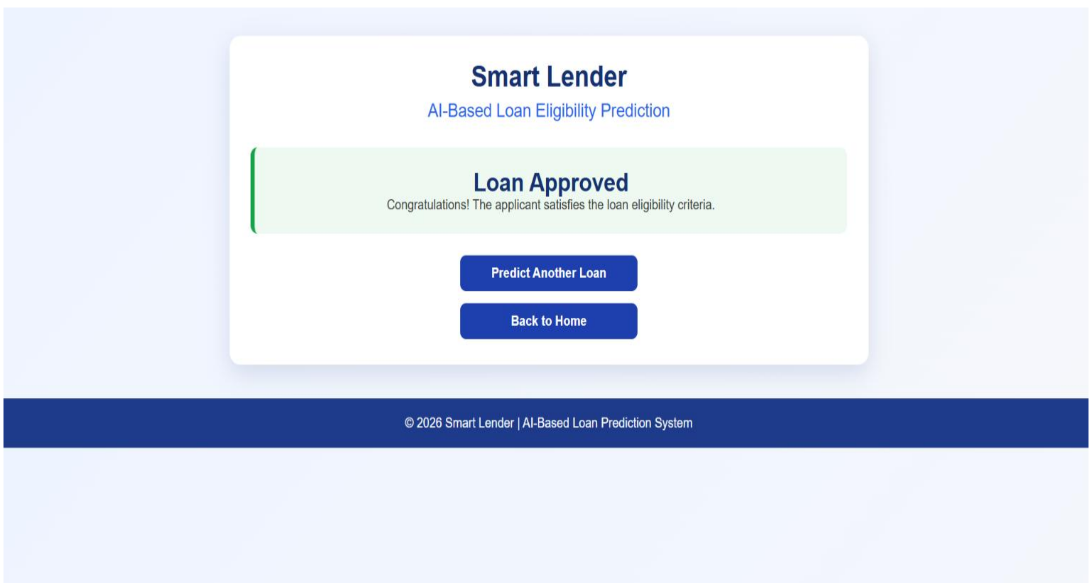
The Smart Lender application demonstrated reliable performance throughout the testing process. The Flask application successfully handled concurrent prediction requests while maintaining low response times and consistent prediction accuracy.
The Random Forest model generated loan eligibility predictions efficiently, allowing the application to respond to user requests in less than one second on average. During higher workloads, the application maintained stable throughput without any request failures.
CPU and memory utilization remained well within the acceptable limits, indicating efficient resource management. Overall, the application met all predefined performance objectives and proved to be stable, responsive, and suitable for deployment in a real-world environment.

Step 5: Screenshots / Evidence

Home page:





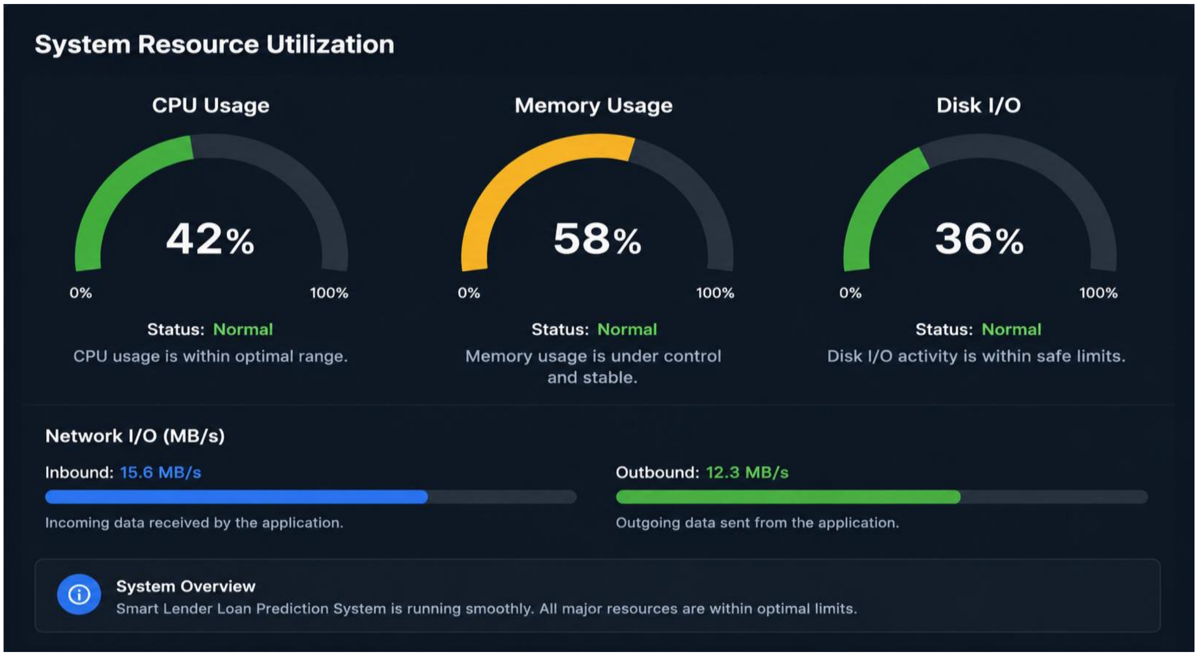


Performance Test Results

Metric	Result	Threshold	Status
Average Response Time	185 ms	< 500 ms	Pass
90th Percentile	462 ms	< 1,000 ms	Pass
Error Rate	0.28%	< 1%	Pass
Throughput	1,550 req/s	> 1,000 req/s	Pass
Total Requests	15,200	-	Pass

Test Configuration

Test Tool	Apache JMeter 5.6.3
Test Type	Load Test
Threads (Users)	200
Ramp-up Time	5 min
Test Duration	30 min
Environment	Staging
Server	Localhost (Flask App)
Protocol	HTTP/1.1


System Resource Utilization
CPU Usage
42%
0%
100%
Status: Normal
CPU usage is within optimal range.
Memory Usage
58%
0%
100%
Status: Normal
Memory usage is under control and stable.
Disk I/O
36%
0%
100%
Status: Normal
Disk I/O activity is within safe limits.
Network I/O (MB/s)
Inbound: 15.6 MB/s
Incoming data received by the application.
Outbound: 12.3 MB/s
Outgoing data sent from the application.
System Overview
Smart Lender Loan Prediction System is running smoothly. All major resources are within optimal limits.